STATS 200A Lecture 8

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$$\int g\,d\mu = \int gf\,d\nu \qquad \text{for measurable} g \geq 0, \text{ or } g \in L^1(\mu)$$

$$\mu(A) = \int_A \frac{1}{\sqrt{2\pi}} e^{-x^2/2}\,\frac{dx}{d\lambda(x)}$$

Definition 8.1
$\mu << \nu$ (*absolutely continuous*) if $\nu(A) = 0 \Rightarrow \mu(A) = 0$

Theorem 8.2 (Radon-Nikodym)
μ, ν σ-finite, $\mu << \nu \Rightarrow \mu = f\nu$ for a measurable $f \geq 0$, unique ν-almsot everywhere. Write $f = \frac{d\mu}{d\nu}$.

Example.

$$\mu([0,a]) = \int_{[0,a]} f(x)\,d\nu(a) \qquad \text{with } f(x) = 2x, \nu = \text{Unif}[0,1]$$

Remark (Change of Measure)**.**

$$\mathbb{P}_1 << \mathbb{P}_0 : \mathbb{E}_{\mathbb{P}_1}[h] = \mathbb{E}_{\mathbb{P}_0}\left[h\frac{d\mathbb{P}_1}{d\mathbb{P}_0}\right] \qquad (h \geq 0)\text{measurable, or } h \in L^1(\mathbb{P}_1)$$

$$\int h\,d\mathbb{P}_1 = \int h\frac{d\mathbb{P}_1}{d\mathbb{P}_0}\,d\mathbb{P}_0$$

$$\mathbb{E}\left[h(x)\right], x \sim \mathbb{P}_1 = \mathbb{E}\left[h(z)\frac{d\mathbb{P}_1}{d\mathbb{P}_0}(z)\right], z \sim \mathbb{P}_0$$

$$\int g\,d\mu = \int gf\,d\nu$$

$$\mu = f\nu \quad (d\mu = f\,d\nu)$$

$$\mathbb{P}_1 = X_{\#}\mathbb{P}$$

Theorem 8.3
There is a unique measure $\mu_1 \otimes \mu_2$ on $\mathcal{E}_1 \otimes \mathcal{E}_2$ with $(\mu_1 \otimes \mu_2)(A \times B) = \mu_1(A)\mu_2(B)$

Theorem 8.4 (Tonelli; Fubini)

On σ-finite measure spaces, let f be product-measurable. If $f \geq 0$ (Tonelli), or $f \in L^1(\mu_1 \otimes \mu_2)$ (Fubini):

$$\int f \, d(\mu_1 \otimes \mu_2) = \int \int f(x,y) \, d\mu_2(y) d\mu_1(x) = \int \int f(x,y) \, d\mu_1(x) d\mu_2(y).$$

Tonelli allows $+\infty$. For Fubini, sections are integrable almost everywhere; all three integrals are finite.

Proof.

(1) **Section measurability.** C_x is measurable, and $x \mapsto \mu - 2(C_x)$ is measurable by $\pi - \lambda$ theorem.

(2) **Construct the product measure.** $(\mu_1 \otimes \mu_2)(C) = \int \mu_2(C_x) \, d\mu_1(x)$.

(3) **Tonelli for indicators.** The reverse-order construction agrees on rectangles; uniqueness $(\pi - \lambda)$ gives the same measure.

(4) **Tonelli for measurable $f \geq 0$.** The standard machine: simple functions, then MCT on the inner and outer integrals.

(5) **Fubini.** Apply Tonelli to $|f|$, f^+ , and f^-: integralable sections almsot everywhere, then subtract the two finite integrals.

■

Theorem 8.5 (Tail Formula)

$X \geq 0, p > 0 : \mathbb{E}X = \int_0^\infty \mathbb{P}(X > t) \, dt, \quad \mathbb{E}X^p = \int_0^\infty pt^{p-1} \mathbb{P}(X > t) \, dt.$

Proof. $X(\omega) = \int_0^\infty \mathbf{1}\left[t < X(\omega)\right] \, dt$. Integrate in ω, swap by Tonelli. The set $\{(\omega, w) : X(\omega) > t\}$ is measurable. ■

Definition 8.6

(X, Y) with joint density f with respect to $\lambda \otimes \lambda \Rightarrow X$ has density $f_X(x) = \int f(x,y) \, dy$

Definition 8.7 (Independent)

σ-algebras $\mathcal{G}_1, \ldots, \mathcal{G}_n \subseteq \mathcal{F}$ are *independent* if $\mathbb{P}\left(\bigcap_{i=1}^n A_i\right) = \mathbb{P}_{i=1}^n \mathbb{P}(A_i)$ for all $A_i \in \mathcal{G}_i$. Random elements X_i are independent if the $\sigma(X_i)$ are. Infinite familes: every finite subfamilies

Theorem 8.8 (check on π-systems)

$\mathcal{P}_i$ π-systems generating $\mathcal{G}_i$. If the product formula holds for all $A_i \in \mathcal{P} \cup \{\Omega\}$, then the $\mathcal{G}_i$ are independent.

Theorem 8.9

$X_1, \ldots, X_n$ are independent if and only if $\mathbb{P}_{(X_1, \ldots, X_n)} = \mathbb{P}_{X_1} \otimes \cdots \otimes \mathbb{P}_{X_n}$.

Proof. independence says the probability measures agree on rectangles, a generating π-system. Apply $\pi - \lambda$. ■

Consequences: Apply pushforward and Tonellie/Fubini:

$$\mathbb{E}\left[\prod_i g_i(X_i)\right] = \prod_i \mathbb{E}[g_i(X_i)]$$

for measurable $g_i \geq 0$, or $\mathbb{E}(g_i(X_i)) < \infty$ for all i.

$$F(x_1, \ldots, x_n) = \prod_i F_i(x_i).$$